Does a learned tokenizer help a small DNA model?

GBST + bidirectional BERT (ChaRNABERT-style), pretrained on the human genome and compared with Transformer++

Omar R.A Mohamed · October 2026 · code, models and results: github.com/OMAR622-ai/dna-xlstm-charna (branch charna-gbst)

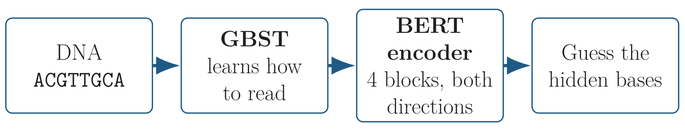
1 The question

Most DNA language models fix in advance how a sequence is cut into "words": single bases, codons or k-mers. **ChaRNABERT** (Morales-Pastor et al., 2024) instead lets the model *learn* how to cut, separately at every position (**GBST**), and reported better results on RNA.

We ask: does this learned reading also help a *small* model of *human DNA*?

Approach: build GBST-BERT in the DNA-xLSTM code base, train it with the DNA-xLSTM protocol (Schmidinger et al., 2024), and compare it with the same-size **Transformer++** that reads one base at a time.

2 The model: GBST-BERT



GBST at every position lists all blocks of 1–4 bases that contain it (10 candidates, like reading frames), scores them, and mixes them with learned weights. Example for the first `T` of `ACGTTGCA`:

Size	Candidate blocks
1	`T`
2	`TT`, `GT`
3	`TTG`, `GTT`, `CGT`
4	`TTGC`, `GTTG`, `CGTT`, `ACGT`

Encoder: attention with RoPE and QKNorm, SwiGLU, LayerNorm (ChaRNABERT §3.3). **Size:** 2.1 M parameters (256 per base, 8 heads), as the 2 M DNA-xLSTM models.

3 Data and training

Human genome hg38 (3.03 billion bases), split as in DNA-xLSTM:

Split	Unique bases	Share
Training	2.33 B	77%
Validation	0.29 B	10%
Test (never trained on)	0.27 B	9%

The game (masked language modelling): 15% of the bases are hidden; the model must guess them from both sides. Half of the time the reverse-complement strand is shown.

```
Real:   A C G T T G C A A G T C
Input:  A C ? T T G C A A ? T C
Answer: .   .   G .   .   .   .   G .   .
```

Setting	Value [source]
Steps	10,000 = 10.5 B bases [Bio-xLSTM A4]
Batch	1,024 seq. × 1,024 bases per step
Optimizer	AdamW (0.9, 0.95), wd 0.1, cosine
LR GBST-BERT	2e-3 (best of 4, our check)
LR Transformer++	1e-2 [Bio-xLSTM Table A3]
Hardware	4 GPUs (JKU student01 / 07), fp32

4 How we test

Loss: how wrong the guesses are: average of $-\ln$(probability of the true base). Lower is better; random guessing = **1.386**.

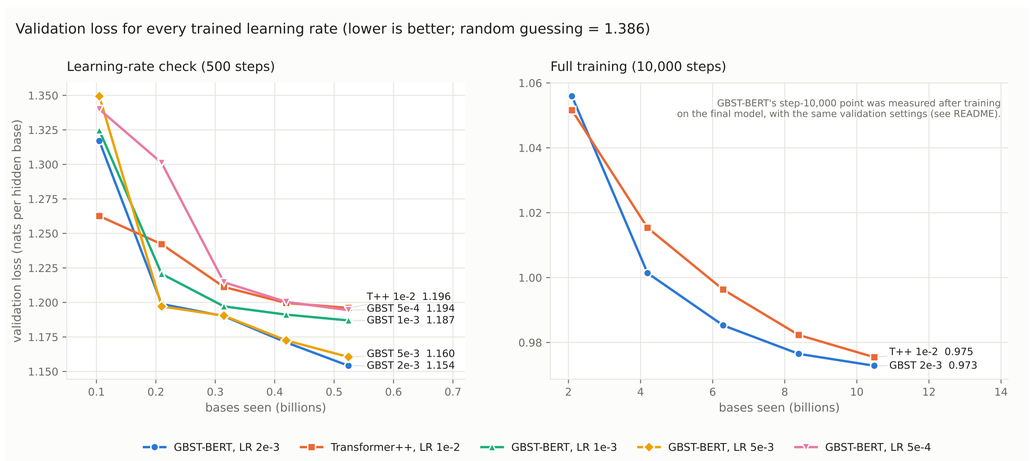
Accuracy: share of hidden bases whose best guess is right, on **2,048 unseen test sequences** (311,767 hidden bases). Both models get *exactly the same* hidden positions.

References without a model: random guessing 25%; always guessing the most common base (A or T) 29.4%.

5 Engineering notes

- 24 automatic tests; checkpoints every 500 steps; every run documented on GitHub.
- **Bug found and fixed** in the repository's genome reader: parallel reader processes shared one open file and sometimes read the wrong bases (0.35% with 2 readers, up to 33% with 12). Fixed with a test (841/1,024 wrong before, 0 after); runs resumed from a clean checkpoint. Test DNA never leaked into training.
- Training moved between servers mid-run and resumed exactly from a checkpoint.

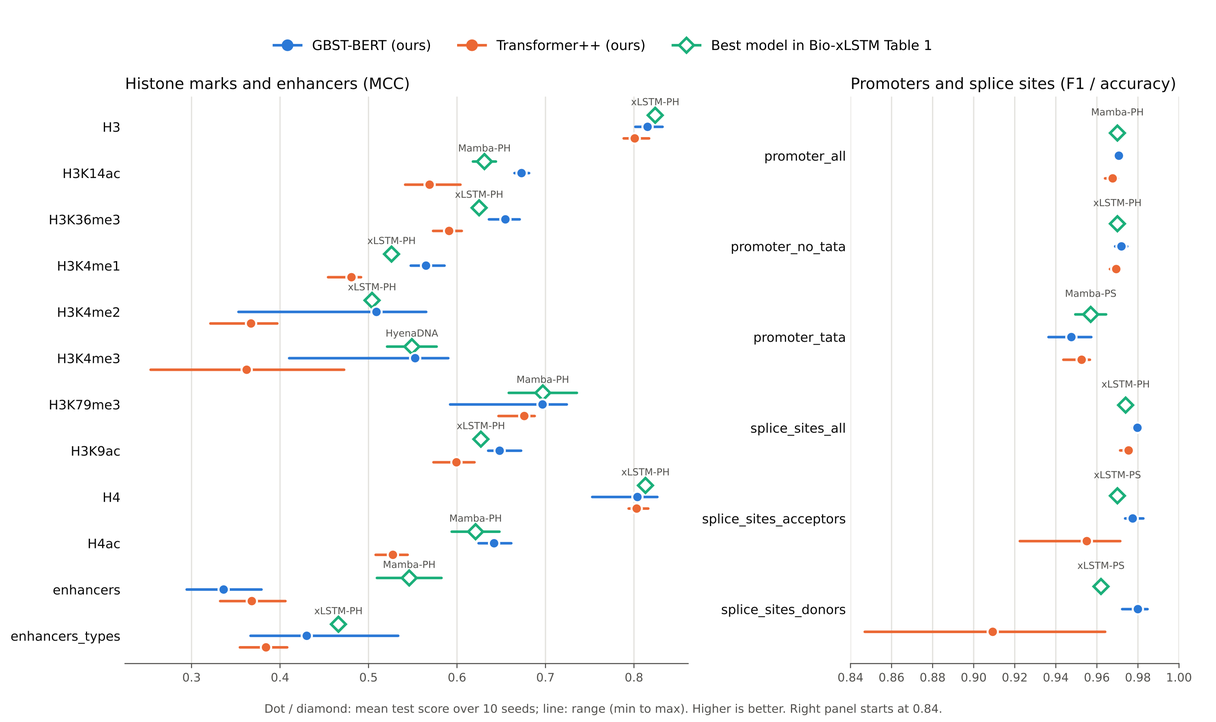
6 Pretraining: validation loss for every learning rate, and the final test



	Model	Learning rate	Steps trained	Val. loss at 0.5 B bases	Val. loss at 10.5 B bases	Final test loss	Final test accuracy	Note
●	GBST-BERT	2e-3	500 + 10,000	1.154	0.973	**0.9753**	**56.44%**	best in the check; used for the full run
◆	GBST-BERT	5e-3	500	1.160	–	–	–	second best
▲	GBST-BERT	1e-3	500	1.187	–	–	–	
▼	GBST-BERT	5e-4	500	1.194	–	–	–	ChaRNABERT paper value (Table 2)
■	Transformer++	1e-2	500 + 10,000	1.196	0.975	**0.9774**	**56.33%**	comparison model; Bio-xLSTM value (Table A3)
	Random guessing			1.386	1.386	1.386	25.00%	reference

How to read it: loss = how wrong the guesses of hidden bases are (lower is better); accuracy = share of hidden bases guessed correctly on unseen test DNA.
Left panel: 500-step check of four GBST-BERT learning rates (2e-3 best). Right panel: the full 10.5-billion-base runs, trained only at each model's chosen learning rate.
Result: GBST-BERT 56.44% vs Transformer++ 56.33% test accuracy — a 0.1-point difference, too small to call a winner after one run per model.

7 Prediction: 18 biology tasks (Nucleotide Transformer benchmark, 10 repeats)



Dot / diamond: mean test score over 10 seeds; line: range (min to max). Higher is better. Right panel starts at 0.84.

Both models fine-tuned as in Bio-xLSTM (same settings, 10 repeats, mean and range). **GBST-BERT has the best mean on 13 of 18 tasks** (xLSTM-PH 4) and beats Transformer++ on **16 of 18**. Weak spot: enhancers.

8 What GBST learned

Share of GBST's weight on blocks of 1 / 2 / 3 / 4 bases (8 sequences, different chromosomes):

Model	1 / 2 / 3 / 4 bases
500 steps, LR 5e-4	72 / 22 / 5 / 1%
500 steps, LR 2e-3	98 / 2 / 0 / 0%
Full run, step 2,000	99.8 / 0.1 / 0 / 0%
Final model	**98 / 2 / 0 / 0%**

On human DNA, GBST learned to read **almost only single bases**; each base still sees its neighbours through the convolution before the blocks. GBST-BERT therefore behaves close to a one-base-at-a-time model.

9 Conclusions

- Both 2 M models learned human DNA well: **56% accuracy** (random 25%), test loss **≈0.975**, in the range of the published 2 M models (≈0.96–1.03).
- In pretraining, GBST-BERT finished only **0.1 points** ahead (56.44% vs 56.33%).
- On the **prediction tasks** the difference is clear: GBST-BERT beats Transformer++ on **16 of 18 tasks** and has the best mean of all seven models on **13 of 18**.
- Enhancers stay weak for both our models.

Caveats: one pretraining run per model; only GBST-BERT's pretraining learning rate was tuned; the models also differ in normalization; fine-tuning settings were the paper's, not tuned per model.

10 Next steps

1. **Clean test of GBST:** the same model with GBST replaced by a plain lookup table (`TOKENIZER=embedding`), same settings.
2. Why are enhancers weak? Longer DNA context (beyond 1,024 bases).
3. Repeat pretraining with other seeds; MLM vs UL2.

Code, models, results (private repository)